ResearchTrail: Literature Reading Path Report

Topic: Vision Transformer **Level:** beginner **Goal:** enter the field **Preferred Length:** 10 papers

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Prerequisites

1. Deep Residual Learning for Image Recognition

Authors: Kaiming He, Xiangyu Zhang, Shaoqing Ren et al. **Year:** 2016 | **Citations:** 210000 **Why read:** Background knowledge needed before diving into the specific topic.

2. Attention Is All You Need

Authors: Ashish Vaswani, Noam Shazeer, Niki Parmar et al. **Year:** 2017 | **Citations:** 145000 **Why read:** Background knowledge needed before diving into the specific topic.

3. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

Authors: Jacob Devlin, Ming-Wei Chang, Kenton Lee et al. **Year:** 2019 | **Citations:** 110000 **Why read:** Background knowledge needed before diving into the specific topic.

Conceptual Foundations

4. A Theoretical Framework for Vision Transformer

Authors: W. Martinez, E. Anderson **Year:** 2000 | **Citations:** 519 **Why read:** Introduces the foundational framework that later work builds upon.

5. An Introduction to Vision Transformer: Theory and Principles

Authors: F. Anderson, B. Anderson, D. Taylor **Year:** 2003 | **Citations:** 1107 **Why read:** Introduces the foundational framework that later work builds upon.

6. An Introduction to Vision Transformer: Theory and Principles: Theory

Authors: P. Taylor **Year:** 2006 | **Citations:** 1086 **Why read:** Introduces the foundational framework that later work builds upon.

Core Methods

7. ImageNet Classification with Deep Convolutional Neural Networks

Authors: Alex Krizhevsky, Ilya Sutskever, Geoffrey E. Hinton **Year:** 2012 | **Citations:** 180000 **Why read:** Central methodological contribution with significant impact (PageRank: 0.033).

Bridge Papers

8. Optimization Methods for Vision Transformer: Theory

Authors: N. Lee **Year:** 2014 | **Citations:** 53 **Why read:** Connects different subfields and provides broader context.

Recent Advances

9. Open Problems and Future Directions in Vision Transformer

Authors: E. Martin, J. Moore **Year:** 2025 | **Citations:** 44 **Why read:** Recent paper (2025) at the research frontier (frontier score: 0.562).

10. Emerging Paradigms in Vision Transformer: Theory

Authors: A. Martinez **Year:** 2025 | **Citations:** 55 **Why read:** Recent paper (2025) at the research frontier (frontier score: 0.562).

11. Towards Vision Transformer: Challenges and Opportunities: Theory

Authors: T. Miller, B. Anderson, T. Perez et al. **Year:** 2025 | **Citations:** 27 **Why read:** Recent paper (2025) at the research frontier (frontier score: 0.561).

Research Graph Statistics

- **Communities detected:** 6
- **Community sizes:** [13, 19, 9, 10, 8, 5]

Top papers by PageRank: - An Introduction to Vision Transformer: Theory and Principles (PR: 0.0397, 2003) - ImageNet Classification with Deep Convolutional Neural Networks (PR: 0.0325, 2012) - Attention Is All You Need (PR: 0.0298, 2017)

Top bridge papers: - Deep Learning Approaches for Vision Transformer (BC: 0.0515) - Deep Learning Approaches for Vision Transformer: Theory (BC: 0.0515) - Attention Is All You Need (BC: 0.0507)

Follow-up Questions You Can Ask

- “Add more survey and tutorial papers”
- “Show bridge papers between key communities”
- “Generate a 7-day reading plan”
- “Show only papers after 2020”
- “Which authors should I follow?”
- “Replace theoretical papers with more applied papers”
- “Explain why these papers are foundational”
- “Show me the citation network visualization”